

AI & ML Task 2: Model Comparison on California Housing

Feature Engineering, Model Optimization & Performance Comparison

1. Objective

Improve on the Task 1 baseline by preparing the data properly, training several regression algorithms, and choosing the final model using measurable metrics (RMSE and R^2).

2. Methodology

Data: California Housing dataset (20,640 districts, 8 numeric features; target = median house value in \$100,000s; no missing values).

Split before scaling: 80/20 train-test split (random_state=42). **StandardScaler** was fit on the training set only and then applied to the test set, which prevents data leakage.

Models: Linear Regression (baseline), Ridge Regression (alpha=1.0, L2 regularisation) and Decision Tree Regressor (max_depth=5, captures non-linear patterns).

Metrics: RMSE (lower is better) and R^2 (higher is better) on the unseen test set; train metrics were also computed to check for overfitting.

3. Results

Model	Train RMSE	Test RMSE	Test MAE	Train R2	Test R2
Linear Regression	0.7197	0.7476	0.5345	0.6126	0.5735
Ridge Regression	0.7197	0.7476	0.5345	0.6126	0.5735
Decision Tree	0.6959	0.7242	0.5223	0.6377	0.5997

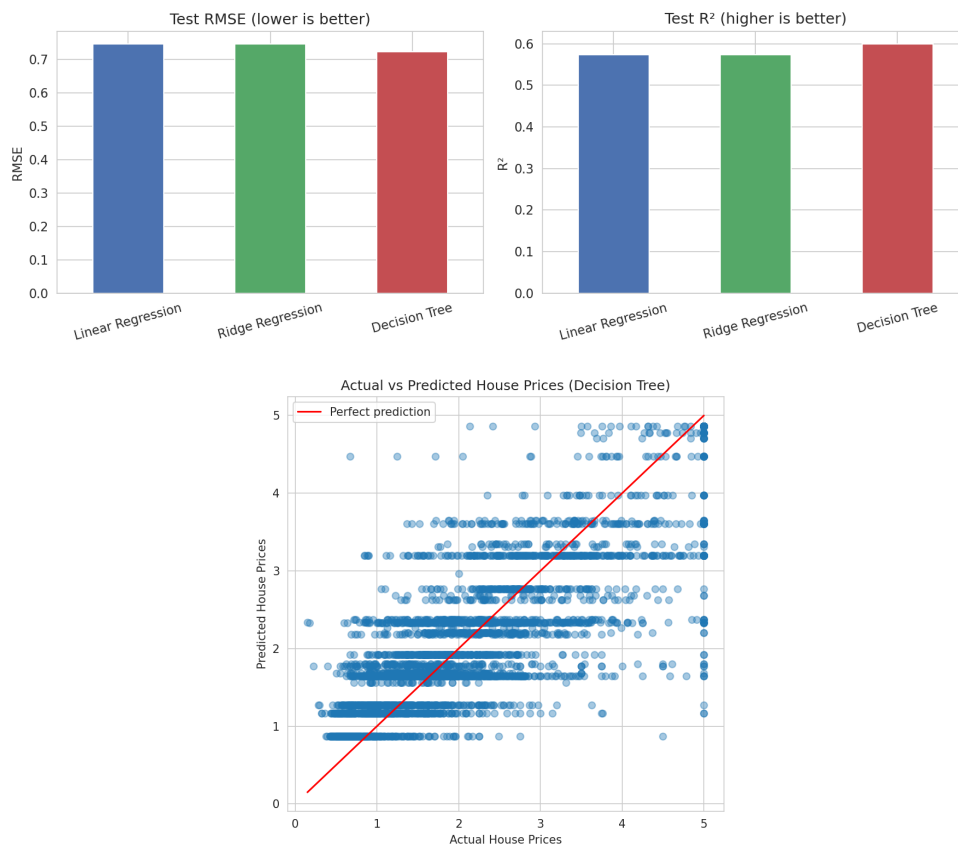


Figure: Actual vs Predicted prices for the selected model. Points closer to the red line mean better predictions.

4. Analysis and Conclusion

Decision Tree was selected because it had the lowest test RMSE (0.7242) and the highest test R^2 (0.5997). Compared with the Linear Regression baseline (RMSE 0.7476, R^2 0.5735), test RMSE dropped by 3.1%. The train-test R^2 gap for the chosen model is 0.038, which is small, so there is no strong sign of overfitting. Ridge Regression matched Linear Regression almost exactly: with thousands of training rows and only 8 features, a small penalty ($\alpha=1.0$) barely changes the coefficients, so the limit is model flexibility (underfitting), not overfitting. The Decision Tree can capture non-linear effects such as the coastal location pattern, which is why it scores higher. A model that always predicts the mean scores $R^2 = -0.00$, so all three models add real signal. The Linear Regression row uses the same 80/20 split ($\text{random_state}=42$) as Task 1, so it is directly comparable to the Task 1 result. Scaling was fit on the training set only to keep the evaluation honest.

Next steps: cross-validation, hyperparameter tuning (GridSearchCV), and ensemble models such as Random Forest or Gradient Boosting (HistGradientBoostingRegressor reached R^2 of about 0.83 in the Task 1 experiment).